

Aegis-Forge I: Thermal Cascade Process Engineering

Stage-by-Stage Temperature, Chemistry, Materials Science, and Control Engineering

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1. Thermal Cascade Engineering — The Science of Sequential Separation

The Aegis-Forge I exploits a fundamental principle of material science: every substance has a discrete phase transition temperature. By raising the drum's internal temperature in precisely controlled increments — the thermal cascade — each material class becomes liquid, decomposes, or volatilizes in a predictable sequence, enabling clean physical separation.

1.1 The Cascade Temperature Ramp Protocol

The complete thermal cycle runs 3.5-5 hours per batch depending on feedstock:

TIME (min)	TEMP (°C)	STAGE	ACTIVITY
0-15	20-80	Pre	Acoustic mineral liberation (Stage 0)
15-35	80-150	1	Moisture removal, volatile organics
35-65	150-450	2	Thermoplastic flow, polymer separation
65-95	450-660	3	Pyrolysis, char formation, thermoset combustion
95-120	420-900	4	Low-melting metals: Pb, Sn, Zn (overlap range)
120-145	660-720	5	Aluminum separation (near-pure Al window)
145-175	720-1085	HOLD	Solidus zone: brass, bronze separation
175-205	1085-1200	6	Copper and copper alloys
205-240	1200-1600	7	Iron, steel, nickel, cobalt
240-270	1600-2200	8	Refractory mineral oxide residue
270-300	2200-cool	Cool	Controlled cool under N2; drum stops rotating

1.2 Stage 0 — Acoustic Pre-Liberation (Ambient)

Before any heating, the feedstock undergoes acoustic pre-treatment in a separate pre-chamber connected to the drum.

Physics of acoustic mineral liberation:

Rock matrix minerals are held together by: 1. Grain boundary adhesion (intergranular cements, silicates) 2. Mechanical interlocking of irregular grain shapes 3. Surface tension of pore fluids

Low-frequency acoustic waves (20-500 Hz) induce cyclic stress waves through the rock matrix. At resonant frequencies specific to the mineral grain size, stress concentrations at grain boundaries exceed the grain boundary fracture toughness:

$$\sigma_{boundary} = \rho \cdot 4\pi^2 f^2 A r^3 K_{IC} > 1 \Rightarrow \text{extliberation}$$

Where A is acoustic amplitude, rho is grain density, r is grain radius, and K_IC is grain boundary fracture toughness. Liberation occurs when the stress ratio exceeds 1.

Low-frequency application (20-200 Hz) — rock matrix loosening:

KNbO3-BaTiO3 piezoelectric transducers mounted on the pre-chamber outer wall. At 100 Hz, 500W continuous: produces standing waves through a 20L charge of gravel. After 15 minutes: mineral liberation index increases from ~20% to ~70% for typical basalt.

Ultrasonic application (20-40 kHz) — fine grain liberation:

Eight 40 kHz ultrasonic transducers (KNbO₃-BaTiO₃ probes, 500W each = 4 kW total) submersed in water bath within pre-chamber. Acoustic cavitation liberates mineral grains at micron scale for REE mineral liberation from host rock.

Bubble collapse pressure in ultrasonic cavitation:

$$P_{collapse} = P_0 \left(\frac{R_0}{R_{min}} \right)^{3\kappa} \approx 10^3 \text{ bar locally}$$

This microsecond, localized pressure is sufficient to fracture silicate-silicate grain boundaries (fracture toughness 0.8-2 MPa·m^{0.5}) without destroying the mineral crystals themselves (harder, more homogeneous materials).

Output of Stage 0: - Rock feedstock disaggregated by 60-80% compared to input - Mineral grains partially liberated from matrix - Downstream thermal cascade is more efficient (larger surface area exposed to induction)

2. Stage 1 — Drying and Volatiles Removal (80-150°C)

2.1 Target Materials

- Free surface water (T_b = 100°C at 1 atm)
- Crystal-bound water in hydrated minerals (releases at 100-200°C depending on mineral)
 - Gypsum (CaSO₄·2H₂O): dehydrates at 107-120°C
 - Borax (Na₂B₄O₇·10H₂O): dehydrates at 100-140°C
 - Kaolin (Al₂Si₂O₅(OH)₄): partially dehydrates at 150°C
- Low-boiling organics: alcohols, acetone residues in electronics cleaning agents
- Solvents from PCB flux (terpineol T_b=218°C, begins at 150°C)

2.2 Temperature Control in Stage 1

Induction heating not effective at 80-150°C (most targets are non-conductive). Alternative heating method: infrared irradiation from SiC glow bars positioned in the static heating elements at drum top position.

SiC glow bars: silicon carbide resistive elements (CSMFAB000000000002-type) — same material used in Ancient Hearth stove burner. Surface temperature: 800-900°C; emitted IR at ~3.6 μm wavelength — absorbed by water strongly (water absorption peak: 3.0 and 6.3 μm).

Drum rotation during Stage 1: 5-10 RPM (slow, gentle tumbling — uniform drying) **Atmosphere:** N₂ at 1.2 bar (prevents explosive H₂/organic vapor combinations) **Duration:** 20 minutes (sufficient for 50 kg charge with <30% moisture content)

2.3 Water and Volatile Recovery

Exiting vapor (mostly H₂O) exits through Port 1 (150°C set-point): - Condensed in BFRP heat exchanger (chilled to 10°C by PEX-a cooling circuit) - Liquid water collected: analyzed for metal content, either purified and reused or discharged - Organic vapors: trace quantity passes through activated carbon bed (char from Stage 3!) — elegant closed cycle: char from Stage 3 later used to adsorb Stage 1/2 organics

3. Stage 2 — Thermoplastic Polymer Flow (150-450°C)

3.1 Target Materials and Their Separation Temperatures

Polymer	Melting/Flow Onset (°C)	Example Sources in Feedstock
Low-Density Polyethylene (LDPE)	105-115°C	Plastic bags, film, coating
High-Density Polyethylene (HDPE)	120-140°C	Pipe, bottles, containers
Polypropylene (PP)	130-171°C	Automotive interior, packaging
Polystyrene (PS)	100-240°C flash	Electronics housings
PVC (polyvinyl chloride)	160-210°C	Wire insulation, pipes
ABS	200-250°C	Electronics housings, dashboards
Nylon PA6	220-230°C	Mechanical parts, connectors
Nylon PA66	255-265°C	Engine compartment parts
PETE/PET	250-260°C	Bottles, film
Polycarbonate (PC)	267-290°C	Lenses, electronic covers
PEEK (CSM own off-cuts)	341°C	Reclaim PEEK from failed parts
PTFE (Teflon)	327°C	Wire insulation, gaskets

Combined melt window: 105°C to 450°C — achievable in single stage by slow ramp.

3.2 Melt Centrifuge Physics in Stage 2

As polymers melt they become viscous (viscosity η at melt: 10^2 - 10^4 Pa·s depending on polymer). Centrifugal force drives molten polymer outward. At 50 RPM ($G=1.4$) low centrifugal forces allow polymers to pool but not fully stratify.

At 150 RPM ($G=12.6$) — adequate for polymer-polymer density separation: - HDPE melt density: 0.78 g/cm^3 - PEEK melt density: 1.22 g/cm^3 - PP density: 0.74 g/cm^3

Density difference relative to G-force creates stratified annular layers. Heavy polymers (PEEK, PC) form outer annular band. Light polymers (PP, LDPE) form inner annular band.

Port 2 at drum OD (Port open at 200°C): primarily discharges highest-density polymer melt first. As temperature rises through 450°C, lower-density polymers also flow.

Output: Mixed polymer slug exiting Port 2, quenched in BFRP mold. This “mixed polymer block” is a downstream input for: - Elium® reformulation (MMA-compatible melt blending) - BFRP matrix additive (modified impact resistance) - Future CSM product: polymer filament for 3D printing

3.3 PVC Warning — Chlorine Management

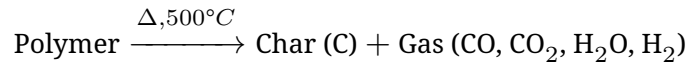
PVC releases HCl vapor at 160-200°C during dehydrochlorination. This is a toxic acid gas. **Critical process control:** - Stage 2 atmosphere: mildly basic N_2 (trace NaHCO_3 aerosol injection — powder spray system) - $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl (salt)} + \text{H}_2\text{O} + \text{CO}_2$: neutralization occurs in-drum - Residual HCl: monitored by inline optical HCl sensor (PMMA fiber + absorption spectroscopy) - If HCl concentration $> 50 \text{ ppm}$: drum temperature hold until HCl level drops - NaCl residue: exits as dissolved mineral in Stage 3 char

4. Stage 3 — Pyrolysis and Carbon Char Formation (450-700°C)

4.1 Thermoset and Organic Decomposition

At temperatures above 450°C, thermosetting polymers (epoxy, phenolic glue, FR4 PCB substrate, rubber) undergo thermal decomposition (pyrolysis):

Simplified global pyrolysis reaction:



Carbon yields from pyrolysis (by weight): - Epoxy resin: 15-25% char yield - FR4 substrate (glass-epoxy): 40-50% char (from glass fiber mass) - Rubber/elastomer: 25-40% char - Phenolic resin: 50-60% char (high carbon yield) - Cellulosic materials: 20-30% char

Temperature details for FR4 PCB substrate: T_{onset}: 350-420°C (depends on halogen content) T_{peak} weight loss: 480-520°C (main decomposition) Residue at 700°C: ~20% (SiO₂ glass fibers + inorganic fillers + carbon char)

4.2 Char Collection — The Valuable Carbonaceous Product

At 600-700°C, char from pyrolysis: - Carbon content: 80-95% C (high after partial oxidation/volatilization) - Surface area: 200-800 m²/g (activated carbon-like porosity from volatile evolution) - Specific resistance: 10⁻³ to 10⁻¹ Ohm·m - Practical applications: 1. Activated carbon for organic vapor adsorption (used immediately in Stage 1 gas treatment!) 2. Carbon black substitute for rubber compounding 3. Graphite precursor for further purification 4. Carburizing agent for steel (reduces iron oxide) 5. Carbon electrodes for electrochemical processes

Char exits Port 3: A dry, granular discharge collected in a ZrO₂-lined ceramic bin. Weight: approximately 8-15% of original organic input.

4.3 Combustion Gas Management

At 500-700°C with residual oxygen: full combustion produces CO₂ + H₂O. Under N₂ atmosphere (oxygen-starved): pyrolysis gases include CO, CH₄, H₂, BTX aromatics.

CSM Operating choice: Maintain <2% O₂ in drum atmosphere (N₂ sweep maintains this). Operate in partial pyrolysis mode □ higher char yield, lower gas yield. Combustible pyrolysis gases (CO, CH₄, H₂) routed to gas management system (Document 111). These gases represent recoverable energy: calorific value ~18-22 MJ/m³ (comparable to town gas).

5. Stage 4 — Low-Temperature Metals (420-900°C): Pb, Sn, Zn

5.1 Target Metal Separation

Metal	Melting Point	Typical Sources	Expected Purity (Raw)
Lead (Pb)	327.5°C	Lead solder, batteries, wheel weights	85-95% Pb
Tin (Sn)	231.9°C	Solder (Sn60/Pb40), tin plate, PCBs	30-60% Sn (alloyed)
Sn-Pb eutectic (63/37)	183°C	Electronic solder joints	63% Sn / 37% Pb
Bismuth (Bi)	271.5°C	Low-temp fusible alloys, cosmetics	variable
Indium (In)	156.6°C	ITO glass coatings, LCD panels	very dilute

Metal	Melting Point	Typical Sources	Expected Purity (Raw)
Zinc (Zn)	419.5°C	Die-cast housings, galvanized steel	85-95% Zn
Zinc alloy (Zamak)	380-390°C	Die-cast automotive, hardware	Zn/Al/Cu alloy

Critical separation for CSMFAB: Indium at 156.6°C. At its melting point, only In liquefies — all other metals remain solid. The ultra-low-temp In separation produces an indium-rich liquid at Port 4 (set to 160°C for In pass).

Process innovation — Indium First Strategy: Heat Stage 4 slowly: 155°C, 157°C, 160°C — In liquefies immediately. Centrifuge at 400 RPM for 5 minutes at exact 160°C. Port 4 opens: In-rich liquid (mixed with residual organics) discharges. Downstream refining of this stream yields 99.9% In via electrodeposition — feeds YInMn production!

This makes the Aegis-Forge I a **domestic Indium recovery machine** — reducing critical mineral dependency.

5.2 Induction Selectivity for Stage 4 Metals

At 420-660°C (Zn melting range): - Zinc: $\rho_{\text{Zn}} = 5.96 \times 10^{-8} \text{ Ohm}\cdot\text{m}$ □ excellent induction coupling - Lead: $\rho_{\text{Pb}} = 2.0 \times 10^{-7} \text{ Ohm}\cdot\text{m}$ □ good induction coupling - At this stage, aluminum is still solid ($T_m = 660^\circ\text{C}$) and provides the “solid matrix”

Induction power (50 kHz, 80 kW): specifically heats Zn/Pb/Sn particles within the solid Al/Cu/Fe matrix. The melt pools at Zn/Pb grain locations, coalesces under G-force, and drains outward.

6. Stage 5 — Aluminum Separation (660-720°C)

6.1 The Aluminum Window

660°C is the liquidus of pure aluminum. Copper melts at 1085°C, iron at 1538°C. The 60°C window between 660°C and 720°C creates a clean aluminum separation stage: - Only Al, Al alloys (6061: 652°C, 7075: 477-635°C solidus), and Al-Si die cast (577°C eutectic) melt - All copper, iron, nickel, cobalt remain solid - Zinc ($T_m=419^\circ\text{C}$) has already been removed in Stage 4

Centrifuge parameters for Stage 5: - Drum speed: 300 RPM ($G=25$) - Aluminum melt viscosity at 700°C: 1.3 mPa·s (extremely fluid — flows easily) - Centrifugal separation is rapid: 5-minute dwell at 680°C to collect >95% of Al content

Expected aluminum slug composition: - Al: 85-97% (depends on input alloy diversity) - Si: 1-8% (from die-cast A380 / A356) - Cu: 0.5-3% (from 2024, 7075 alloys) - Mg: 0.3-1.5% This mixed alloy slug goes to downstream refining (Document 112) for property-specific casting.

CSMFAB relevance: Recovered aluminum not used directly in Aegis products (which avoid metals), but sold as commercial scrap to fund operations, or used in non-structural tooling (molds, fixtures).

7. Stage 6 — Copper Separation (1085-1200°C)

7.1 Copper Recovery

At 1085°C, copper melts. Copper is present in: - PCB traces and plated-through holes (20-35% Cu by weight in PCB) - Motor windings (99.9% Cu) - Plumbing scrap (99.5% Cu pipe) - Electrical wire (99.9% Cu)

Induction heating of copper at Stage 6: Coil frequency: 5 kHz (penetration depth in Cu at 1100°C: ~4mm — suitable for wire/trace sizes) Power: 150-200 kW to overcome Cu's high heat capacity ($0.385 \text{ J/g}\cdot\text{K} \times 200 \text{ kg} \times 1065^\circ\text{C} = 82 \text{ MJ}$)

Special consideration — Precious metals with copper: Gold ($T_m = 1064^\circ\text{C}$), Silver ($T_m = 961.8^\circ\text{C}$), Platinum ($T_m = 1768^\circ\text{C}$) Both Au and Ag melt before or at Stage 6 temperature. They report with copper in Port 6 output. Downstream electrolytic refining separates these from the Cu slug — this is a REVENUE STREAM.

PCB precious metal content (per ton of PCBs): - Gold: 300-400 g/ton - Silver: 3,000-5,000 g/ton - Palladium: 100-200 g/ton - Copper: 150,000-250,000 g/ton (0.15-0.25 kg/board)

At 2026 gold price \$3,200/troy oz $\square$ 300g/ton Au = **\$30,900/ton PCB in gold alone**

8. Stage 7 — Ferrous Metal Cascade (1200-1600°C)

8.1 Iron, Steel, and Ferromagnetic Metal Recovery

Ferrous metals are uniquely efficiently heated by induction (ferromagnetic cores couple at 2-10 $\times$ the power density of non-ferrous metals). The Stage 7 induction system therefore uses the LOWEST frequency (2 kHz) for maximum penetration and ferromagnetic coupling.

Metal	Melting Point	Induction Coupling
Carbon steel	1370-1400°C	Excellent (ferromagnetic below Curie Temp 770°C)
Stainless (304)	1400-1430°C	Good (austenitic, non-magnetic — eddy current only)
Cast iron	1150-1200°C	Excellent (high carbon, approaches Cu at boundary)
Nickel	1455°C	Very good (ferromagnetic $T_c=358^\circ\text{C}$; non-mag at Stage 7 temps but conductive)
Cobalt	1495°C	Good
Chromium	1907°C	Remains solid — segregates to Stage 8 residue

Iron slug chemical analysis target: Fe: >90%; C: 2-4%; Si: 1-2%; Mn: 0.5-1% This low-alloy iron/steel slug can be remelted for general castings or sold as secondary iron.

Rare earth magnets (NdFeB, SmCo) in Stage 7: NdFeB $T_m = 1177^\circ\text{C}$ $\square$ melts in Stage 7 window. SmCo $T_m = 1490^\circ\text{C}$ $\square$ melts in Stage 7 as well. Combined RE-Fe-B and RE-Co melts report to Port 7 with main ferrous stream. IMPORTANT: This stream contains Nd, Sm, Co — valuable rare earths for new magnet production! Downstream refining (hydromet leach) separates RE from the Fe matrix.

9. Stage 8 — Refractory Mineral and Ceramic Residue (1600-2200°C)

9.1 What Remains at 1600°C+

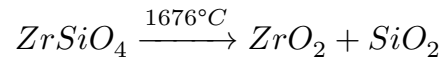
After all organic matter, polymers, metals through iron have been removed, the remaining solids at 1600°C are oxide minerals and refractory ceramics:

- **Silica glass fibers (from FR4 PCBs, glass insulation):** $T_m = 1600-1700^\circ\text{C}$ $\square$ begins melting
- **Zirconia (ZrO₂ from Aegis product returns):** $T_m = 2716^\circ\text{C}$ — solid throughout; concentrated
- **Rare earth oxides (from magnet returns, ore processing):** Y₂O₃, La₂O₃, Nd₂O₃ $\square$ remain solid

- **Titanium dioxide (TiO₂ from ilmenite processing):** decomposes above 2000°C
- **Alumina (Al₂O₃):** T_m = 2072°C — primarily solid in Stage 8; some partial melt
- **Silicon carbide (SiC) residue:** sublimation at 2700°C — solid through Stage 8
- **Zircon sand (ZrSiO₄):** dissociates at 1676°C → ZrO₂ + SiO₂ (valuable dissociation!)

9.2 Zircon Sand Processing — Critical CSMFAB Material Recovery

At 1700°C, zircon sand (ZrSiO₄) dissociates:



The ZrO₂ is then available for: - Further processing to 3Y-TZP ceramic powder (add Y₂O₃, re-mill, sinter)
- ZrB₂ synthesis (add B₂O₃ + carbon → carbothermal reduction)

The Stage 8 mineral residue that exits Port 8 is: A partially fused slag-like material rich in Zr, Ti, rare earth oxides, Al₂O₃, SiO₂. This is the primary input for the Materials Study Part V critical mineral recovery program.

Output: “Mineral Concentrate Slug” — the treasure chest Chemical analyses of Port 8 output from mixed electronics recycling (per ton input): - ZrO₂: 0.5-3% (from ITO-bearing LCD glass) - TiO₂: 1-4% (from white pigments in plastics, ilmenite-bearing minerals) - Rare Earth Oxides: 0.1-2% (from magnets, phosphors, catalysts) - Al₂O₃: 15-25% (from anodized Al oxide, aluminosilicates) - SiO₂: 40-60% (glass, PCB substrate) - Carbon (unoxidized): 2-5% This slag feeds the Phoenix Protocol hydrometallurgical refinery.

*End of CSMFAB000000000109 Volume 2 | Thermal Cascade Process Engineering Continued in Volume 3:
Acoustic Mineral Liberation Subsystem*